

**Question 1:**

Which StringBuilder variable fails to compile?

```
public class StringBuilderInstantiations {  
    public static void main( String[] args ) {  
        var stringBuilder1= new StringBuilder();  
        var stringBuilder2 = new StringBuilder(10);  
        var stringBuilder3 = new StringBuilder("Java");  
        var stringBuilder4 = new StringBuilder(new char[]{'J','a','v','a'});  
    }  
}
```

- A. stringBuilder4
- B. stringBuilder3
- C. stringBuilder1
- D. stringBuilder2
- E. None of them

**Question 2:**

Consider the following methods to load an implementation of `MyService` using `ServiceLoader`. Which of the methods are correct? (Choose all that apply)

- A. `MyService service = ServiceLoader.services(MyService.class).getFirstInstance();`
- B. `MyService service = ServiceLoader.load(MyService.class).findFirst().get();`
- C. `MyService service = ServiceLoader.load(MyService.class).iterator().next();`
- D. `MyService service = ServiceLoader.getService(MyService.class);`

**Question 3:**

Given:

1. `Map<String, Integer> map = Map.of( "b", 1, "a", 3, "c", 2 );`
2. `TreeMap<String, Integer> treeMap = new TreeMap<>( map );`
3. `System.out.println( treeMap );`

What is the output of the given code fragment?

- A. {c=2, a=3, b=1}
- B. {c=1, b=2, a=3}
- C. {a=1, b=2, c=3}
- D. {a=3, b=1, c=2}
- E. {b=1, a=3, c=2}
- F. {b=1, c=2, a=3}

**Question 4:**

Given:

```
class A implements Serializable {  
    int number = 1;  
}
```

```
class B implements Serializable {  
    int number = 2;  
}
```

```
public class Test {  
    public static void main( String[] args ) throws Exception {  
        File file = new File( "o.ser" );  
        A a = new A();  
        var oos = new ObjectOutputStream( new FileOutputStream( file ) );  
        oos.writeObject( a );  
  
        var ois = new ObjectInputStream( new FileInputStream( file ) );  
        B b = ( B ) ois.readObject();  
        System.out.println( b.number );  
    }  
}
```

What is the given program's output?

- A. ClassCastException
- B. 1
- C. 2
- D. NotSerializableException
- E. Compilation fails

**Question 6:**

How would you create a ConcurrentHashMap configured to allow a maximum of 10 concurrent writer threads and an initial capacity of 42?

Which of the following options meets this requirement?

- A. `var concurrentHashMap = new ConcurrentHashMap();`
- B. `var concurrentHashMap = new ConcurrentHashMap(42);`
- C. `var concurrentHashMap = new ConcurrentHashMap(42, 0.88f, 10);`
- D. None of the suggestions.
- E. `var concurrentHashMap = new ConcurrentHashMap(42, 10);`

**Question 7:**

Given:

```
var frenchCities = new TreeSet<String>();  
frenchCities.add("Paris");  
frenchCities.add("Marseilsle");  
frenchCities.add("Lyon");  
frenchCities.add("Lille");  
frenchCities.add("Toulouse");  
System.out.println(frenchCities.headSet( "Marseille" ));
```

What will be printed?

- A. Compilation fails
- B. [Paris]
- C. [Paris, Toulouse]
- D. [Lyon, Lille, Toulouse]
- E. [Lille, Lyon]

**Question 8:**

Given:

```
public class Test {  
    static int count;  
  
    synchronized Test() {  
        count++;  
    }  
  
    public static void main( String[] args ) throws InterruptedException {  
        Runnable task = Test::new;  
        Thread t1 = new Thread( task );  
        Thread t2 = new Thread( task );  
        t1.start();  
        t2.start();  
        t1.join();  
        t2.join();  
        System.out.println( count );  
    }  
}
```

What is the given program's output?

- A. Compilation fails
- B. It's always 2
- C. It's either 0 or 1
- D. It's always 1
- E. It's either 1 or 2

**Question 9:**

Given:

1. `System.out.print(Boolean.logicalAnd( 1==1,2<1 ));`
2. `System.out.print(Boolean.logicalOr( 1==1,2<1 ));`
3. `System.out.print(Boolean.logicalXor( 1==1,2<1 ));`

What is printed?

- A. `true true false`
- B. `false true true`
- C. Compilation fails
- D. `true false true`
- E. `true true true`



**Question 11:**

Given:

```
1. interface SmartPhone {  
2.     boolean ring();  
3. }  
4.  
5. class Iphone15 implements SmartPhone {  
6.     boolean isRinging;  
7.  
8.     boolean ring() {  
9.         isRinging = !isRinging;  
10.        return isRinging;  
11.    }  
12. }
```

Choose the right statement.

- A. An exception is thrown at running Iphone15.ring();
- B. SmartPhone interface does not compile
- C. Everything compiles
- D. Iphone15 class does not compile

**Question 12:**

Given:

```
1. Optional o1 = Optional.empty();  
2. Optional o2 = Optional.of( 1 );  
3. Optional o3 = Stream.of( o1, o2 )  
4.                  .filter( Optional::isPresent )  
5.                  .findAny()  
6.                  .flatMap( o -> o );  
7. System.out.println( o3.orElse( 2 ) );
```

What is the given code fragment's output?

- A. Optional[1]
- B. An exception is thrown
- C. 0
- D. Optional.empty
- E. 1
- F. Compilation fails
- G. 2

**Question 13:**

What do the following print?

```
1. public class DefaultAndStaticMethods {
2.     public static void main( String[] args ) {
3.         WithStaticMethod.print();
4.     }
5. }
6.
7. interface WithDefaultMethod {
8.     default void print() {
9.         System.out.print( "default" );
10.    }
11.}
12.
13.interface WithStaticMethod extends WithDefaultMethod {
14.    static void print() {
15.        System.out.print( "static" );
16.    }
17.}
```

- A. default
- B. Compilation fails
- C. nothing
- D. static

### Question 14:

Given:

```
1. Period p = Period.between(  
2.     LocalDate.of( 2023, Month.MAY, 4 ),  
3.     LocalDate.of( 2024, Month.MAY, 4 ) );  
4. System.out.println( p );  
5. Duration d = Duration.between(  
6.     LocalDate.of( 2023, Month.MAY, 4 ),  
7.     LocalDate.of( 2024, Month.MAY, 4 ) );  
8. System.out.println( d );
```

What is the output?

(A)

1. PT8784H
2. UnsupportedOperationException

(B)

1. P1Y
2. PT8784H

(C)

1. P1Y
2. UnsupportedOperationException

(D)

UnsupportedTemporalTypeException

(E)

1. PT8784H
2. P1Y

**Question 15:**

Given:

```
1. List integers = List.of( 0, 1, 2 );  
2. integers.stream()  
3.     .peek( System.out::print )  
4.     .limit( 2 )  
5.     .forEach( i -> {  
6.         } );
```

What is the output of the given code fragment?

- A. 01
- B. 012
- C. An exception is thrown
- D. Compilation fails
- E. Nothing

**Question 17:**

Given:

```
1. StringBuilder result = Stream.of( "a", "b" )
2.                               .collect(
3.                               () -> new StringBuilder( "c" ),
4.                               StringBuilder::append,
5.                               ( a, b ) -> b.append( a )
6.                               );
7. System.out.println( result );
```

What is the output of the given code fragment?

- A. bac
- B. cab
- C. acb
- D. cbca
- E. abc
- F. cba
- G. cacb
- H. bca

**Question 18:**

What do the following print?

```
1. public class Main {  
2.  
3.     int instanceVar = staticVar;  
4.     static int staticVar = 666;  
5.  
6.     public static void main( String args[] ) {  
7.         System.out.printf( "%d %d", new Main().instanceVar, staticVar );  
8.     }  
9.  
10.    static {  
11.        staticVar = 42;  
12.    }  
13.}
```

- A. Compilation fails
- B. 42 42
- C. 666 666
- D. 666 42

**Question 19:**

Given:

```
1. public class Test {  
2.  
3.     class A {  
4.     }  
5.  
6.     static class B {  
7.     }  
8.  
9.     public static void main( String[] args ) {  
10.        // Insert here  
11.    }  
12. }
```

Which three of the following are valid statements when inserted into the given program?

- A. A a = new Test().new A();
- B. B b = new Test.B();
- C. A a = new Test.A();
- D. B b = new B();
- E. B b = new Test().new B();
- F. A a = new A();



**Question 21:**

Given:

```
1. void verifyNotNull( Object input ) {  
2.     boolean enabled = false;  
3.     assert enabled = true;  
4.     assert enabled;  
5.     System.out.println( input.toString() );  
6.     assert input != null;  
7. }
```

When does the given method throw a NullPointerException?

- A. A NullPointerException is always thrown if the input argument is null
- B. Only if assertions are disabled and the input argument is null
- C. Only if assertions are enabled and the input argument isn't null
- D. Only if assertions are enabled and the input argument is null
- E. A NullPointerException is never thrown
- F. Only if assertions are disabled and the input argument isn't null

**Question 22:**

Which of the following isn't a correct way to write a string to a file?

(A)

```
1. try ( BufferedWriter writer = new BufferedWriter( "file.txt" ) ) {  
2.     writer.write( "Hello" );  
3. }
```

(B)

None of the suggestions

(C)

```
1. Path path = Paths.get( "file.txt" );  
2. byte[] strBytes = "Hello".getBytes();  
3. Files.write( path, strBytes );
```

(D)

```
1. try ( PrintWriter printWriter = new PrintWriter( "file.txt" ) ) {  
2.     printWriter.printf( "Hello %s", "James" );  
3. }
```

(E)

```
1. try ( FileOutputStream outputStream = new FileOutputStream( "file.txt" ) )  
    {  
2.     byte[] strBytes = "Hello".getBytes();  
3.     outputStream.write( strBytes );  
4. }
```

(F)

```
1. try( FileWriter writer = new FileWriter( "file.txt" ) ){  
2.     writer.write("Hello");  
3. }
```

**Question 24:**

Given:

```
1. String textBlock = ""  
2.           j \  
3.           a \t  
4.           v \s  
5.           a \  
6.           "";  
7. System.out.println(textBlock.length());
```

What is the output?

- A. 12
- B. 10
- C. 11
- D. 14

**Question 26:**

Given:

```
1. Deque<Integer> deque = new ArrayDeque();
2. deque.offer( 1 );
3. deque.offer( 2 );
4. var i1 = deque.peek();
5. var i2 = deque.poll();
6. var i3 = deque.peek();
7. System.out.println( i1 + " " + i2 + " " + i3 );
```

What is the output of the given code fragment?

- A. 1 2 1
- B. An exception is thrown
- C. 1 1 1
- D. 2 2 2
- E. 2 2 1
- F. 2 1 2
- G. 2 1 1
- H. 1 2 2
- I. 1 1 2

**Question 30:**

Given:

```
1. public class Test {  
2.     public static void main( String[] args ) throws IOException {  
3.         Path p1 = Path.of( "f1.txt" );  
4.         Path p2 = Path.of( "f2.txt" );  
5.         Files.move( p1, p2 );  
6.         Files.delete( p1 );  
7.     }  
8. }
```

In which case does the given program throw an exception?

- A. An exception is always thrown
- B. File f1.txt exists while file f2.txt doesn't
- C. Both files f1.txt and f2.txt exist
- D. Neither files f1.txt nor f2.txt exist
- E. File f2.txt exists while file f1.txt doesn't

**Question 37:**

What do the following print?

```
1. import java.util.stream.Stream;
2.
3. public class StreamReduce {
4.     public static void main( String[] args ) {
5.         Stream<String> stream= Stream.of("J","a","v","a");
6.         System.out.print(stream.reduce( String::concat ));
7.     }
8. }
```

- A. Compilation fails
- B. Optional[Java]
- C. null
- D. Java

**Question 38:**

Which two of the following aren't the correct ways to create a Stream?

- A. `Stream<String> stream = Stream.builder().add( "a" ).build();`
- B. `Stream stream = Stream.ofNullable( "a" );`
- C. `Stream stream = Stream.of( "a" );`
- D. `Stream stream = Stream.empty();`
- E. `Stream stream = new Stream();`
- F. `Stream stream = Stream.of();`

**Question 40:**

Given:

1. `StringBuffer us = new StringBuffer( "US" );`
2. `StringBuffer uk = new StringBuffer( "UK" );`
3. `Stream<StringBuffer> stream = Stream.of( us, uk );`
4. `String output = stream.collect( Collectors.joining( "-", "=", "" ) );`
5. `System.out.println( output );`

What is the given code fragment's output?

- A. Compilation fails
- B. `=US-UK`
- C. `-US=UK`
- D. `US-UK`
- E. `US=UK`
- F. An exception is thrown



**Question 46:**

Given:

```
1. interface A {  
2.     default void ma() {  
3.     }  
4. }  
5.  
6. interface B extends A {  
7.     static void mb() {  
8.     }  
9. }  
10.  
11. interface C extends B {  
12.     void ma();  
13.  
14.     void mc();  
15. }  
16.  
17. interface D extends C {  
18.     void md();  
19. }  
20.  
21. interface E extends D {  
22.     default void ma() {  
23.     }  
24.  
25.     default void mb() {  
26.     }  
27.  
28.     default void mc() {  
29.     }  
30. }
```

Which interface can be the target of a lambda expression?

- A. B
- B. C
- C. A
- D. D
- E. E
- F. None of the above

**Question 49:**

Given:

```
1. class MyResource implements AutoCloseable {  
2.  
3.     public void close() throws Exception {  
4.         throw new IOException();  
5.     }  
6. }  
7.  
8. public class Test {  
9.     public static void main( String[] args ) {  
10.         try ( MyResource resource = new MyResource() ) {  
11.             throw new RuntimeException();  
12.         } catch ( Exception e ) {  
13.             System.out.println( e );  
14.         }  
15.     }  
16. }
```

What is the given program's output?

- A. java.io.IOException
- B. java.lang.RuntimeException
- C. Compilation fails
- D. java.lang.Exception
- E. Nothing

**Question 50:**

Given:

1. `Stream<String> strings = Stream.of("United", "States");`
2. `BinaryOperator<String> operator = (s1,s2)->s1.concat(s2.toUpperCase());`
3. `String result = strings.reduce("-", operator);`
4. `System.out.println(result);`

What is the output of this code fragment?

- A. United-States
- B. United-STATES
- C. -UnitedStates
- D. UnitedStates
- E. -UNITEDSTATES
- F. UNITED-STATES
- G. -UnitedSTATES